RflySim evidence navigator

Measured 10 October 2026. Offline component handoff 11 October.
Branch audit/rflysim. Source revision 9749a40. No new simulator run. AirGPU kept Off.

Component decision

Reuse PX4/CopterSim for stock motion and control integration. Keep native RGB as a visual baseline. Two flights passed on fresh PX4 estimates. An external Gaussian renderer can use exported motion and define its own camera/time, after the selected state-frame mapping is verified.

Native calibrated replay remains incomplete. Direct photo/Gaussian import is unresolved. Reconstructed-mesh import is documented but Free build eligibility is unverified. An external visual-controller loop would require substantial integration. No real reconstruction, training dataset or learned weights entered this evaluation.

Standalone review files

rflysim-component-handoff-20261011.pptx - 2,368,298 bytes, private Drive download hash verified.

rflysim-component-handoff-20261011.pdf - 4,228,850 bytes, private Drive download hash verified.

Detailed report / Installed interfaces / Exact inventory

Playable native videos

A1 video - 260 frames, 26.0 s at fixed 10 FPS. Invalid intended rotations.

A2 video - 261 frames, 26.1 s at fixed 10 FPS. Invalid intended rotations.

B2 video - 365 frames, 36.5 s at fixed 10 FPS. Concurrent RGB during waypoint control.

Every video includes warmup and all saved frames. Ordinal playback does not use acquisition time. A1 capture was 25.797 s, below its 26 s contract, despite 26.0 s playback. Original 640x480 PNGs are lossless evidence. Videos add a 40 px caption below the image.

Existing private audit folder - permissions unchanged. The Markdown navigator and exact JSON inventory are in this same folder.

Results and raw archive locator

Test	Outcome and measurement basis
B1, no camera	Minimum + full route PASS. 38.750 s supervised. PX4 estimates, fresh mode/arming/landed state.
B2, RGB camera	Minimum + full route PASS. 39.219 s supervised, 365 frames. Images did not drive control.
A1 / A2	260 / 261 frames. Intended rotations invalid through our helper defect. Native geometric checks incomplete.
Rate	351 frames / 35 s including 5 s warmup, 300 post-warmup. Delivered record rate 10.013 Hz.
Coupled / C	Concurrent B2 recording demonstrated, acquisition-pose association blocked. C did not run.

Archives and exact entry names

Checkpoint 1
Renderer logs, stock stdout/stderr, prerequisite record, probes1-2 and early code.

Checkpoint 2 / B1
B1/flight.jsonl, run.json, config.json, freeze.json and empty cleanup.jsonl. B1 GPU/resources, stdout/stderr, probes3-5.

Main raw archive / camera + B2
B2/flight.jsonl + run/config/freeze/empty cleanup. B2/rgb/camera.jsonl and frames (365). A1 and A2 camera.jsonl/frames (260/261), rate camera.jsonl/frames (351). B1-native.ulg, probes6-10, reset records and immutable code-v2.

Original review bundle
Historical 10 October report/deck and selected telemetry/plots/code. Revised standalone files supersede its narrative.

Native B1 ULog: 51,318,508 bytes, 946.224 s of startup plus flight. Offline parsing recovers firmware identity, 660 initial PX4 parameters and 11 changes. No corruption/dropout records reported. This log contains PX4 estimates, not dynamics ground truth.

Mac locations, verification and gaps

Exact local base:

/Users/srreyanshsethi/Downloads/AL/R4R-Sims-rflysim

Output	Location below this base
Current report/deck/PDF	results/rflysim/eval-20261010T215531Z/README.md, rflysim-evaluation.pptx, rflysim-evaluation.pdf
Flight records	artifacts/rflysim/eval-20261010T215531Z/raw/B1/flight.jsonl and B2/flight.jsonl (20,050 / 20,280 entries)
Native PNGs + clocks	raw/A1/, A2/, rate/, B2/rgb/ beneath that same artifact root. Each has frames/ and camera.jsonl. 1,237 PNGs total.
Derived records/plots	results/rflysim/eval-20261010T215531Z/B1/ and B2/: actions, estimates, ACK/mode/arming, truth diagnostics, trajectories/timing/resources.
B1 ULog	artifacts/rflysim/eval-20261010T215531Z/raw/B1-native.ulg
Videos	artifacts/rflysim/handoff-20261011T220420Z/videos/
Navigator and exact hashes	results/rflysim/handoff-20261011T220420Z/README.md, outputs.json, videos.json, drive-publication.json, local-source-verification.json
Original report/deck	Git9749a40 and artifacts/rflysim/handoff-20261011T220420Z/before/

Verification

Three ZIP hashes and 27 + 33 + 1,300 internal entries verify on the Mac. All 1,237 PNGs open. MP4 decode counts match source counts. New standalone videos, PPTX and PDF match full private Drive downloads by size/SHA256. Executed raw/code snapshots remain separate from later corrections.

Remaining unknowns

Final transcript, B2 ULog, strict-duration A1 retest files and final scene/model/launcher/B2 parameter inventory remain unexported. Native projection/extrinsics/optical axes and acquisition association are unverified. Raw cleanup files are empty; workers hit 15 s despite confirmation summaries. Flight records retain landing evidence. GPU B2 coverage is partial; renderer-only throughput/minimum hardware are unknown. Real captures/reconstruction/held-out views and downstream evaluation inputs are absent.

Smallest future action

No restart is needed for the component verdict. Optional recovery-only session: about 15-20 min, no reinstall or rerun. External motion mapping: another 10-15 min. Native corrected A: 10-15 min after startup, projection/extrinsics 15-25 min, clock association 10-20 min. Cleanup retest: 5-10 min only if reusing the helper. These overlap and are planning estimates. Preserve all numerical criteria and do not automatically repeat passed B. See docs/rflysim/minimal-followup.md.